

Evolutionary and functional study of the CDPK gene family in wheat (*Triticum aestivum* L.)

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Abstract Calcium-dependent protein kinases (CDPKs) are crucial sensors of calcium concentration changes in plant cells under diverse endogenous and environmental stimuli. We identified 20 CDPK genes from bread wheat and performed a comprehensive study on their structural, functional and evolutionary characteristics. Full-length cDNA sequences of 14 CDPKs were obtained using various approaches. Wheat CDPKs were found to be similar to their counterparts in rice in genomic structure, GC content, subcellular localization, and subgroup classification. Divergence time estimation of wheat CDPK gene pairs and wheat–rice orthologs suggested that most duplicated genes already existed in the common ancestor of wheat and rice. The number of CDPKs in diploid wheat genome was estimated to be at least 26, a number close to that in rice, *Arabidopsis*, and poplar. However, polymorphism among

EST sequences uncovered transcripts of all three homologous alleles for 13 out of 20 CDPKs. Thus, the hexaploid wheat should have 2–3 fold more CDPK genes expressing in their cells than the diploid species. Wheat CDPK genes were found to respond to various biotic and abiotic stimuli, including cold, hydrogen peroxide (H₂O₂), salt, drought, powdery mildew (*Blumeria graminis tritici*, *Bgt*), as well as phytohormones abscisic acid (ABA) and gibberellic acid (GA). Each CDPK gene often responded to multiple treatments, suggesting that wheat CDPKs are converging points for multiple signal transduction pathways. The current work represents the first comprehensive study of CDPK genes in bread wheat and provides a foundation for further functional study of this important gene family in *Triticeae*.

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Abbreviations

ABA	Absciscic acid
<i>Bgt</i>	<i>Blumeria graminis tritici</i>
CDPK	Calcium-dependent protein kinase
GA	Gibberellin
H ₂ O ₂	Hydrogen peroxide
MYA	Million years ago
NLS	Nuclear localization signal

Introduction

Sessile plants have developed sophisticated signaling pathways to deal with dramatic environmental changes that may affect their normal growth, such as pathogen attack,

drought, and cold. Calcium is a universal second messenger that responds to these stimuli. Transient changes in calcium concentration in the cytoplasm are sensed and decoded by an array of calcium sensors, such as calmodulins, calmodulin-like proteins, calcineurin B-like proteins and calcium-dependent protein kinases (Snedden and Fromm 1998; Trewavas and Malho 1998; Snedden and Fromm 2001; Reddy 2001; Sanders 2002; Luan et al. 2002). Calcium-dependent protein kinases (CDPKs), with its functional kinase domain coupled with regulatory calcium-binding EF-hands, have been identified throughout the plant kingdom from algae to angiosperms (Ludwig et al. 2004). Other than plants, CDPKs are only found in some protozoans (Harmon et al. 2001; Harper and Harmon 2005; Ishino et al. 2006).

CDPKs possess a variable N-terminal domain and several functional domains, including a protein kinase domain, an autoinhibitory region and a calmodulin-like domain (Harper et al. 1991; Harmon et al. 2001; Cheng et al. 2002; Hrabak et al. 2003). The N-terminal domain is highly variable and often bares myristoylation or palmitoylation sites that are believed to be associated with subcellular targeting (Cheng et al. 2002). CDPKs are known to be directly activated by calcium binding to their calmodulin-like domain, which causes conformational change in the autoinhibitory domain that exposes active sites on the kinase domain (Reddy 2001; Sanders 2002). Activated CDPKs participate in different signal transduction pathways through phosphorylation of specific substrates (Zhang and Chollet 1997; Hernandez et al. 2004). In addition to calcium, CDPKs are also regulated by phospholipids (Szczegielniak et al. 2000, 2005) and 14-3-3 proteins (Camoni et al. 1998; Cheng et al. 2002).

CDPKs have been demonstrated to be involved in regulating plant responses to a variety of stimuli, including hormones, cold/drought/salt stress, elicitor, light and sucrose (Urao et al. 1994; Botella et al. 1996; Sheen 1996; Frattini et al. 1999; Patharkar and Cushman 2000; Saijo et al. 2000; Romeis et al. 2001; Murillo et al. 2001; Yang and Komatsu 2001; Chico et al. 2002; Martínez-Noël et al. 2007; Tsai et al. 2007; Wan et al. 2007). In rice, transgenic plants with a disrupted SPK (a seed-specific CDPK) exhibited reduced accumulation of storage starch and proteins in immature seeds (Asano et al. 2002). Transgenic rice constitutively expressing OsCDPK7 and OsCDPK13 showed enhanced tolerance to cold, salt and drought stress (Saijo et al. 2000; Komatsu et al. 2007). In tobacco, CDPK-silenced plants displayed a reduced and delayed hypersensitive response to fungal Avr9 elicitor (Romeis et al. 2001). Heterologous expression of a grape CDPK ACPK1 in Arabidopsis provided evidence that ACPK1 is involved in ABA signal transduction as a positive regulator, suggesting that the ACPK1 gene may be potentially

used for elevating plant biomass production (Yu et al. 2007). Two Arabidopsis guard cell-expressed CDPK genes, CPK3 and CPK6, are shown to be involved in transducing stomatal ABA signaling (Mori et al. 2006). A recent work provided *in planta* genetic evidence for the involvement of CDPK/calcium in ABA signaling at the whole-plant level and showed that CPK4 and CPK11 are two important positive regulators in CDPK/calcium-mediated ABA signaling pathways (Zhu et al. 2007). In addition, the *Medicago truncatula* CPK3 participates in the regulation of the symbiotic interaction (Gargantini et al. 2006). All above evidence clearly demonstrated that the CDPK genes are associated with functions to confer plants with increased resistance to abiotic and biotic stresses. However, the molecular mechanisms of most CDPKs and their target proteins in plants are still not clear.

A genome-wide analysis of Arabidopsis CDPKs identified 34 CDPK genes (Cheng et al. 2002; Hrabak et al. 2003). A similar attempt in rice revealed 29 CDPK genes by using an earlier version of the genome sequence (Asano et al. 2005). Recently, two additional proteins were discovered to belong to this gene family, making the total number of CDPKs in rice to 31 (Ray et al. 2007). Both Arabidopsis and rice CDPKs can be divided into four distinct subgroups. In Arabidopsis, the 34 CDPK genes are distributed among all five chromosomes. However, five CDPK genes, AtCPK31, AtCPK27, AtCPK22, AtCPK21 and AtCPK23 are located in tandem on chromosome 4. These CDPKs all belong to group II-a on the phylogenetic tree, suggesting they arose by gene duplication events (Cheng et al. 2002; Hrabak et al. 2003). In contrast, tandem duplication of CDPK genes was not observed in the rice genome (Asano et al. 2005). In rice, phylogenetic analysis based on protein sequences distinguished 11 closely related pairs of rice CDPKs, some of which may have arisen via segmental duplication events.

Despite the extensive studies of CDPKs in many other plants (Cheng et al. 2002; Hrabak et al. 2003; Asano et al. 2005), little is known about this gene family in bread wheat. Till now, only one wheat CDPK gene TaCDPK1 was reported to be involved in sucrose induced signaling pathways in wheat leaves (Martínez-Noël et al. 2007). Moreover, most CDPK genes of wheat have not been clearly correlated with responses to specific biotic and abiotic stresses. Wheat (*Triticum aestivum* L.) is one of the most important sources of protein in human diet. It is a staple for 35% of the world population and supplies approximately 20% of the calories consumed worldwide (<http://www.cymmt.org/>). Losses from adverse environmental changes greatly impact wheat production. Modulation of signaling regulators is one of the most promising approaches for improving stress tolerance of plants. Bread wheat is a hexaploid species with three genomes A, B, and D that were derived from closely related

ancestral genomes. The ancestral species of diploid wheat are considered to be originated at approximately the same time as rice around 50–70 million years ago (MYA) (Wolfe et al. 1989). The direct diploid progenitors of the hexaploid wheat were estimated to have diverged for 2.5–4.5 MYA (Huang et al. 2002). The hexaploid bread wheat itself appeared and then was domesticated only about 8,000 years ago from spontaneous hybridization between a tetraploid species *T. turgidum* and a diploid goat grass *Aegilopes tauschii* (McFadden and Sears 1946; Jaaska 1980; Lubbers et al. 1991; Talbert and Clack 1991; Dvorak et al. 1998). Most bread wheat genes present in the genome as homoeologous triplicates as shown by EST mapping (Qi et al. 2004). Wheat is notorious for its huge genome, which confers a high percentage of repetitive sequences. Sequencing wheat genome does not appear to be realized in the near future.

In wake of the importance of CDPK genes in biotic and abiotic stress responses, we initiated a project to isolate CDPKs from bread wheat. Sequences related to 20 wheat CDPK genes were identified. Full-length cDNA sequences of 14 genes were obtained using various approaches. Evolution history reconstruction via comparison with rice CDPK genes indicated that a diploid wheat genome should have a similar number of CDPKs to rice. However, EST investigation discovered up to three highly similar, but distinguishable transcripts for each CDPK, indicating the expression of three homoeologous copies in this hexaploid species. When subject to various abiotic and biotic stresses, transcriptional level changes were observed for most wheat CDPK genes, suggesting that they participated in stress related signal transduction pathways. The current work represents the first comprehensive study of CDPK genes in bread wheat and provides a foundation for further functional investigation of this important gene family in *Triticeae*.

Materials and methods

Plant materials and treatments

Wheat cv Chinese Spring was grown in a growth chamber at 22°C and a photoperiod of 16 h (60 $\mu\text{mol m}^{-2} \text{s}^{-1}$ photon flux density). For salt, drought, H_2O_2 and ABA treatment, 7-day-old seedlings were transferred to a Hogland solution containing 200 mM NaCl, 16% PEG, 10 mM H_2O_2 or 5 μM ABA, respectively. Root tissues were harvested at 0, 3, 6, 12 and 24 h for salt, drought and H_2O_2 treatment, and at 0, 0.5, 1, 2, 3 h and at 0, 3, 6, 12, 24 h respectively for ABA treatment. After rinsing with ddH₂O three times, the collected samples were blotted dry and stored at –70°C until RNA extraction.

For cold treatment, 7-day-old seedlings were transferred from 22°C to 4°C refrigerator and kept for 24 h. For powdery

mildew infection, powdery mildew conidia (*Blumeria graminis tritici*, Bgt) on heavily diseased plants were shaken off over a settling tower onto the wheat primary leaves. For GA treatment, 5 μM GA were sprayed to the surface of the leaves of 7-day-old seedlings. Leaves were harvested at 0, 3, 6, 12 and 24 h for above three treatments. Collected samples were stored at –70°C until RNA extraction.

RNA isolation and semi-quantitative RT-PCR

Wheat tissues were ground into fine powder in liquid N₂ and total RNA was extracted using TRIZOL reagent following the manufacturer's instructions (Tian-wei-shi-dai, China). First strand cDNA was synthesized using the SuperScript first-strand synthesis kit (Invitrogen, CA, USA). After the cDNA synthesis reaction, RT-PCR was performed (for isolating the full-length cDNAs of CDPK genes) with 35 cycles of 94°C for 30 s, 55°C for 30 s, and 72°C for 45 s. Gene-specific primers for each CDPK were shown in Table S1. For semi-quantitative RT-PCR, tubulin concentration was used to adjust the concentration of cDNA. The same batch of templates were used in the following RT-PCRs. PCR parameters used for tubulin were: 1 min at 94°C, 1 min at 55°C, 1 min at 72°C with a gradual increase of cycles of 28, 30 and 32 cycles. PCRs for CDPK expression assay were generally carried out with 30 cycles of 1 min at 94°C, 1 min at 55°C, 1 min at 72°C, or adjusted to give best differentiation. Each PCR was repeated at least three times.

Plasmid constructs and imaging

Subcellular localization assay was conducted as described by (Wang et al. 2004). Complete coding sequences of seven wheat CDPK genes were amplified using the primers shown in Table S1. PCR products were subcloned into pDONR201 vector to generate entry clones (Invitrogen, BP and LR enzyme kits). Entry clones with full-length cDNAs were recombined with the C-GFP vector to generate pTaCPK-GFP containing a TaCPK-GFP fusion construct under the control of CaMV 35S. Each construct was confirmed by sequencing before used for transient transformation of onion (*Allium cepa*) epidermis cells using a gene gun (Bio-Rad). Transformed onion cells were observed under a confocal microscope (Nikon).

Identification of wheat CDPK genes

The hidden Markov model (Eddy 1998) was exploited for identifying new CDPK genes in wheat. Using HMMER

2.1.1 software package (<http://www.hmmmer.wust.edu>), two profiles were built out of multiple alignments of kinase and EF hand domains of rice CDPKs. These patterns were then used to search for CDPK members among wheat ESTs that were six-frame translated.

Four methods were used to amplify the coding sequences. The first method is using two pairs of nested primers designed for each CDPK genes whose tentative consensus (TCs) contain complete open reading frame (ORF, Table S1). The second method is screening of wheat full-length cDNA library. The third method is using direct PCR amplification from full-length cDNA phage library mixtures of wheat. The fourth method is using 5' and 3' RACE. PCR fragments were cloned into pGEM-T vector (Tianwei-shi-dai, China) and sequenced on a 3730 capillary sequencer (Applied Biosystems). Wheat CDPK sequences reported in this work have been submitted to GenBank under the accession numbers EU181177–EU181193. Functional domains were predicted at the ExPASy website (<http://www.expasy.org/tools/scanprosite/>).

Chromosomal localization and gene structural analysis

To determine the chromosomal location, 21 nulli-tetrasomic (NT) lines of Chinese Spring were used. Gene specific primers for CDPK gene localization were showed in Table S1. UR203 (*T. urartu*, AA genome) and Y2280 (*T. tauschii*, DD genome) genomic DNA were extracted from leaves using the cetyltrimethylammonium bromide procedure (Murray and Thompson 1980). Primers used to amplify the genomic DNA were listed in Table S1. These genomic sequences have been submitted to GenBank with accession numbers EU183487–EU183491 and EU200767.

Phylogenetic analysis and divergence time estimation

Full length protein sequences of 34 Arabidopsis, 29 rice and 14 wheat CDPK proteins were aligned using ClustalW (Thompson et al. 1994). Alignments were adjusted manually in Jalview (<http://www.jalview.org/>, Clamp et al. 2004). A phylogenetic tree was created according to the neighbor-joining method using MEGA3.1 program (Saitou and Nei 1987; Kumar et al. 2004).

Global alignment of CDPK gene pairs were done using DNASTar MegAlign function. Synonymous (K_S) and non-synonymous (K_A) substitution rates were calculated at the PAL2NAL web server (<http://www.coot.embl.de/pal2nal/>, Suyama et al. 2006), which uses the codeml program of PAML (Yang 1997) to estimate K_S and K_A . DNA divergence time of wheat and rice CDPK gene pairs were estimated using the divergence rate $r = 6.5 \times 10^{-9}$

mutations per synonymous site per year (Gaut et al. 1996) and the formula $T = K_S/2r$ (Li 1997).

Results

Isolation of wheat CDPK genes

To isolate wheat CDPK genes, we first translated the sequences in the TIGR wheat gene index in six-frames and searched against them using HMMER2.1.1 program (Eddy 1998). Hidden Markov models were produced using kinase and EF hand domains from rice CDPK proteins. The first batch of full-length wheat CDPK genes were TCs with both kinase and four EF hands. Sequence information from these TCs was applied in full-length cDNA isolation. Second, we compared all wheat or barley ESTs (a total of >1.5 million) with rice CDPK proteins and those with significant similarity were chosen for full-length cDNA isolation. CDPK proteins were often longer than 500 amino acids, corresponding to an ORF of >1500 bp. Additional difficulty in getting full-length cDNAs of wheat CDPKs is that their 5' region often had high GC content, up to 90%. To search for more full-length cDNAs, we also screened the ESTs from our own full-length cDNA libraries (Zhao et al. unpublished) and completely sequenced the putative full-length clones. Additional effort was made for those CDPKs with only 3' sequence information by using 5' and 3' RACE experiments or direct PCR amplification of the ligation products for full-length cDNA library constructions. 5' RACE to obtain the 5' coding region was not successful for CDPKs with high GC content. Since the three homoeologous copies from the A, B, and D genomes were often highly similar, in order to avoid chimeric sequences full-length DNA inserts were usually amplified from cDNA mixtures using primers covering the complete ORF. One clone was sequenced completely, such that each represented a unique CDPK gene. After all aforementioned efforts, we were able to identify 20 unique wheat CDPK genes, of which 14 possessed a complete ORF (Table 1). A 15th full-length sequence was obtained for TaCPK16 from *Ae. tauschii*, but contained a chimeric region in the middle of the protein sequence. The first stop codon was found at the 621th bp from the start codon for TaCPK16. This sequence may represent a degrading pseudogene, if it were not due to our cloning error (data not shown). During this project, nine accessions of wheat CDPK genes appeared in GenBank that are either identical or putative homoeologs of TaCPK1 (ABK23687), TaCPK2 (AY704444), TaCPK4 (DQ333376, DQ333377), TaCPK5 (DQ333374, DQ431850, DQ333375, DQ431851) and TaCPK8 (AJ621356) (Table 1). Recently, one wheat CDPK (AJ621356, identical to TaCPK8 in this study) is reported by Martínez-Noël et al. (2007) and shown to be involved in sucrose induced signaling pathway in

Table 1 Characteristics of CDPKs from wheat

Name	cDNA		Tigr EST/TC		Amino acid (aa)	No. of EF hands	Sub group	No. of alleles	N-terminal aa	Myristoylation motif	Subcellular localization	Chromosome
	Full length	Length (bp)	No	CDS ^a								
TaCPK1	Y	2,153	TC252299 ^b	Y	548	4	I	3	MGNQNGTP	N	Similar to free GFP	5A ^c
TaCPK2	Y	1,694	TC247204 ^b	Y	558	4	I	3	MGNACGGS	Y	Plasma membrane	2AL 2BL 2DL ^d
TaCPK3	Y	1,994	TC270498	N	545	4	IIIb	3	MGNCCRSP	N	Plasma membrane	–
TaCPK4	Y	2,295	TC266250 ^b	Y	514	4	I	3	MQPDASGN	N	Similar to free GFP	4A ^c
TaCPK5	Y	1,961	TC265323 ^b	Y	532	4	IIa	3	MGQCCSRAT	Y	Plasma membrane	2AS 2BS 2DS ^d
TaCPK6	Y	1,554	TC246172	Y	509	4	IV	2	MGLCTSSS	N	–	–
TaCPK7	Y	2,050	TC252008	Y	543	4	IIIb	3	MGNCCATP	N	–	2B ^c
TaCPK8	Y	1,574	TC252083 ^b	Y	548	4	IIb	3	MGNRTSRH	Y	–	4D ^c
TaCPK9	Y	2,280	BE498083	N	562	4	I	2	MGNTCAGP	N	Similar to free GFP	–
TaCPK10	N	1,696	TC255319	N	–	4	IV	3	–	–	–	6AS 6BS 6DS ^d
TaCPK11	N	1,293	TC256213	N	–	4	–	2	–	–	–	6A ^c
TaCPK12	Y	2,214	CV770153	N	535	4	IIIb	2	MGNCCGAP	N	–	5A ^c
TaCPK13	Y	2,258	BJ258339	N	518	4	IIb	3	MGNCCAGS	N	–	–
TaCPK14	N	1,194	CV765059	N	–	–	–	3	–	–	–	1A 1B ^c
TaCPK15	Y	2,221	TC248321	N	551	4	IIIb	3	MGNCCRSP	N	Plasma membrane	1BL ^d
TaCPK16	N	2,017	TC242486	N	–	4	I	3	–	–	–	5A 5B 5D ^c
TaCPK17	N	694	CA744920	N	–	–	–	1	–	–	–	6AL 6BL 6DL ^d
TaCPK18	Y	1,975	CJ614636	N	534	4	IIa	2	MGNCFKA	N	–	–
TaCPK19	Y	2,189 ^c	CJ626979	N	569	4	IIIb	3	MGNACFSC	N	–	–
TaCPK20	N	774	BQ802750	N	–	–	III	1	–	–	–	–

^a Complete coding sequence; ^b GenBank accession ABK23687, AY704444, DQ333376 (DQ333377), DQ431850 (DQ333374, DQ333375, DQ431851) and AJ621356 nearly identical to TC252299, TC247204, TC266250, TC265323 and TC252083; ^c This study; ^d By bin map (Qi et al. 2004); ^e From *T. carthlicum*; –No data available

excised wheat leaves. Its role in stress responses, however, was not studied.

Features of wheat CDPK genes

As mentioned above, CDPK proteins have a variable N-terminal, often containing a myristoylation motif and/or a palmitoylation motif, which may be responsible for membrane association. The full-length wheat CDPK proteins we cloned were 509 to 569 amino acids long and the size variation was largely caused by different lengths of the N termini (Table 1). In total, 12 CDPKs were tested regarding their subcellular localization using onion epidermal cells. Seven TaCPK-GFP constructs provided fluorescence signals as shown in Fig. 1. TaCPK2 and

5 were predicted to have a myristoylation motif (Table 1) and were found to be associated with plasma membranes when expressed in onion epidermal cells (Fig. 1e and g). However, it seems that proteins lacking myristoylation motif can also locate to the plasma membrane since two wheat CDPKs, TaCPK3 and 15 were shown to be membrane-bounded (Fig. 1h and f). This may indicate the existence of additional motifs that provide these CDPKs with their membrane association ability, which probably was not detected by the computer program. TaCPK3 and 15 also contained a bipartite nuclear localization signal (NLS) sequence in their junction domain and their sequence at the second EF hand loop was deviated from the consensus (data not shown). Therefore, TaCPK3 and 15 potentially can migrate from plasma membrane to the nucleus during signal transduction. Three CDPKs

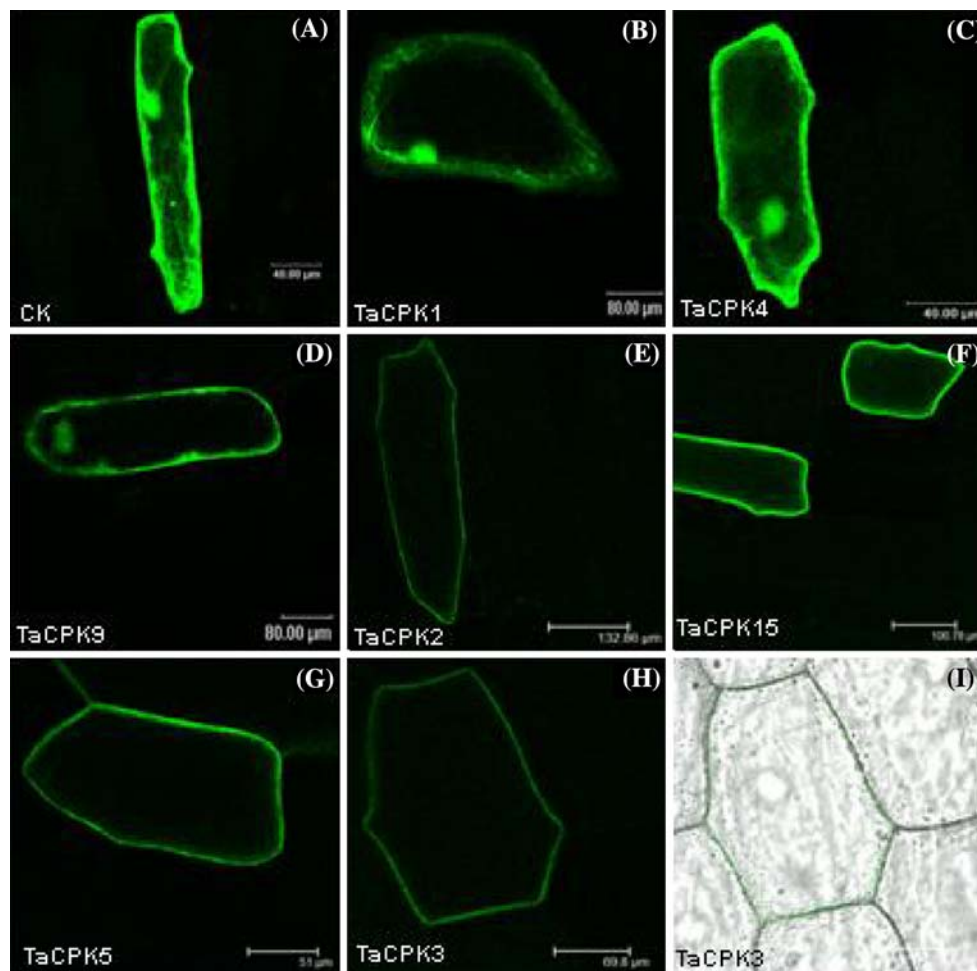


Fig. 1 Subcellular localization of seven wheat CDPK proteins. Confocal images of onion epidermal cells expressing either GFP (a) or TaCPK-GFP fusion proteins (b–h). Panels (b)–(d) show free distribution of TaCPK1, TaCPK4, and TaCPK9. (e)–(h) Show plasma

membrane association of TaCPK2, TaCPK15, TaCPK5, and TaCPK3. Panel (i) is an overlay of TaCPK3-GFP fluorescence and white-field images

(TaCPK1, TaCPK4, and TaCPK9) were found to be located in both the cytosol and the nucleus (Fig. 1b–d), although all three CDPK isoforms do not contain NLS when analyzed by ProSite (Falquet et al. 2002).

All wheat CDPK proteins with full-length sequences had a calmodulin-like structure with four EF hands at their C-terminus (Table 1). We did not observe CDPKs with less than four EF hands as in Arabidopsis and rice. Global alignment showed that sequence identity between most wheat CDPK proteins were below 80%, except TaCPK3 and TaCPK15 whose protein sequences were more than 87% identical (Table S2), suggesting they were derived from recent duplications as will be discussed below. Monocot genes have been shown to have high GC content around their initiation codon region (Wong et al. 2002; Asano et al. 2005). Such GC content variation was also observed in some wheat CDPK genes (Fig. 2). Figure 2a shows the GC content of TaCPK19 that had a high GC content throughout the gene, whereas TaCPK5 was

opposite. For TaCPK7 and TaCPK12, the first portion of the gene had a GC content of above 60%, in contrast to the second half of the gene, which had a GC content of below 50% (Fig. 2b). Interestingly, such GC content distribution was conserved between putative wheat and rice orthologous genes, suggesting GC content may be important for the function of monocot genes.

Sequence comparison of wheat CDPKs with those of rice and Arabidopsis

In order to compare wheat CDPKs with those of rice and Arabidopsis, a phylogenetic tree was generated using full length protein sequences (Fig. 3). Similar to that in rice and Arabidopsis, wheat CDPKs were clearly classified into four major groups. As shown in Fig. 3, Groups II and III comprised of two subgroups, whereas Group IV had the least number of genes. To determine whether the genomic

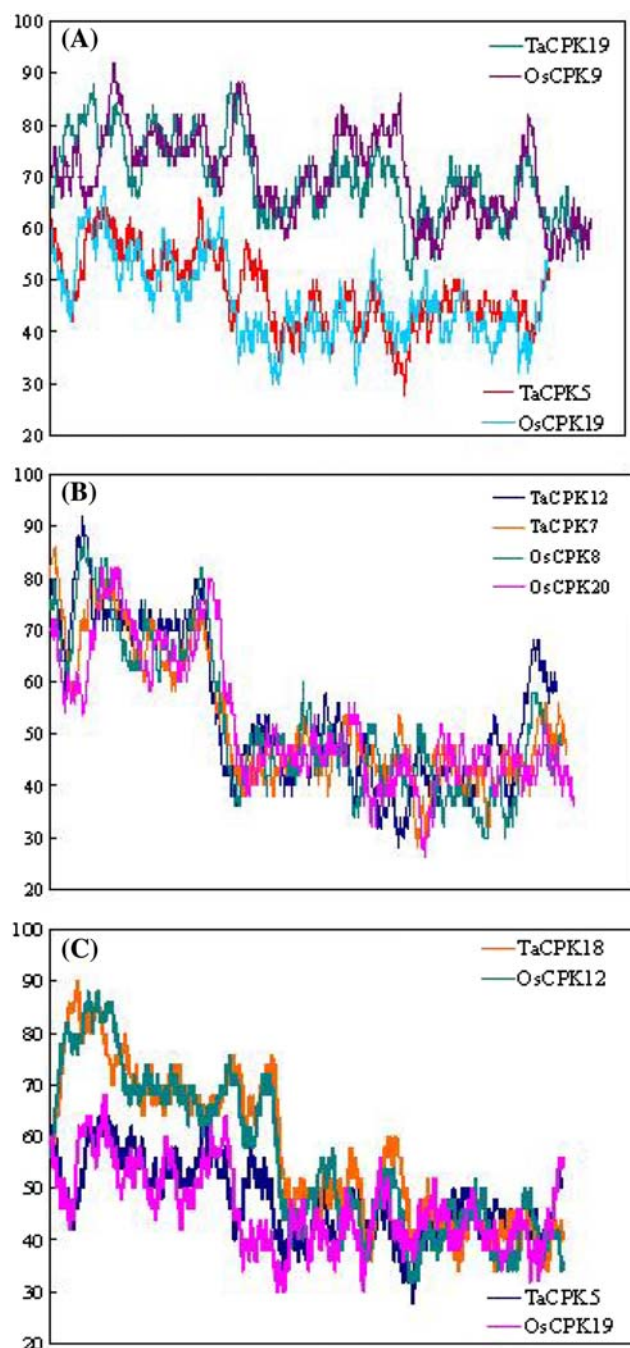


Fig. 2 Comparison of GC contents of representative wheat CDPK genes and their putative rice orthologs. GC contents were analyzed using GENETYX software with a 51 bp sliding window (Asano et al. 2005). (a) The overall high GC content of TaCPK19 and OsCPK9 and the overall low GC content of TaCPK5 and OsCPK19. (b) Contrasting high GC content of the first portion and the low GC content of the second portion of wheat TaCPK7/TaCPK12 and rice OsCPK8/OsCPK20 gene pairs. (c) GC content divergence of putative duplicated CDPK genes of wheat TaCPK5 and TaCPK18 and rice OsCPK19 and OsCPK12

structures of CDPKs are conserved in wheat, we isolated genomic sequences of wheat CDPK gene. To avoid chimeras, genomic DNA of diploid wheat species *T. urartu*

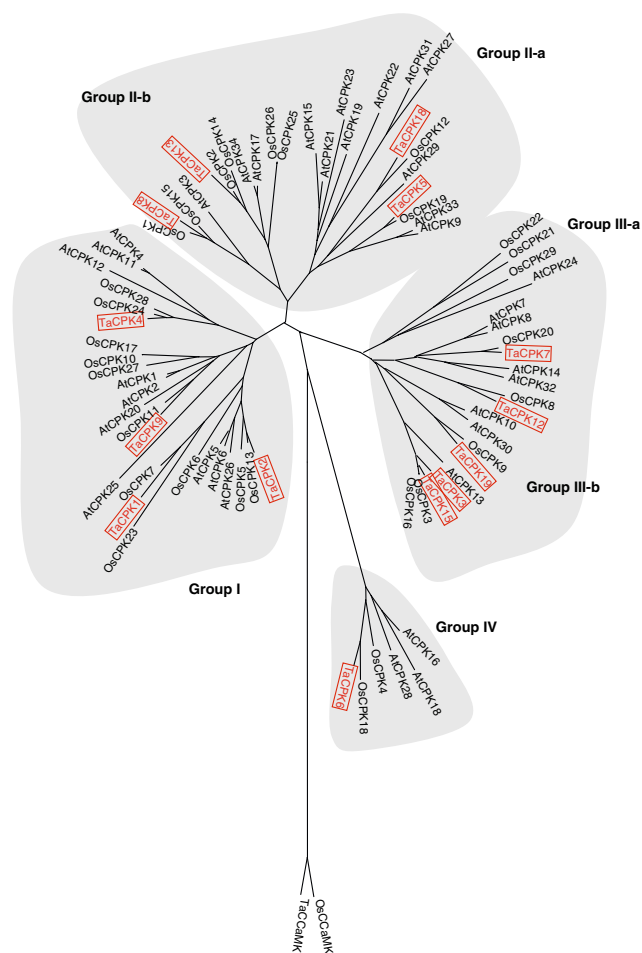
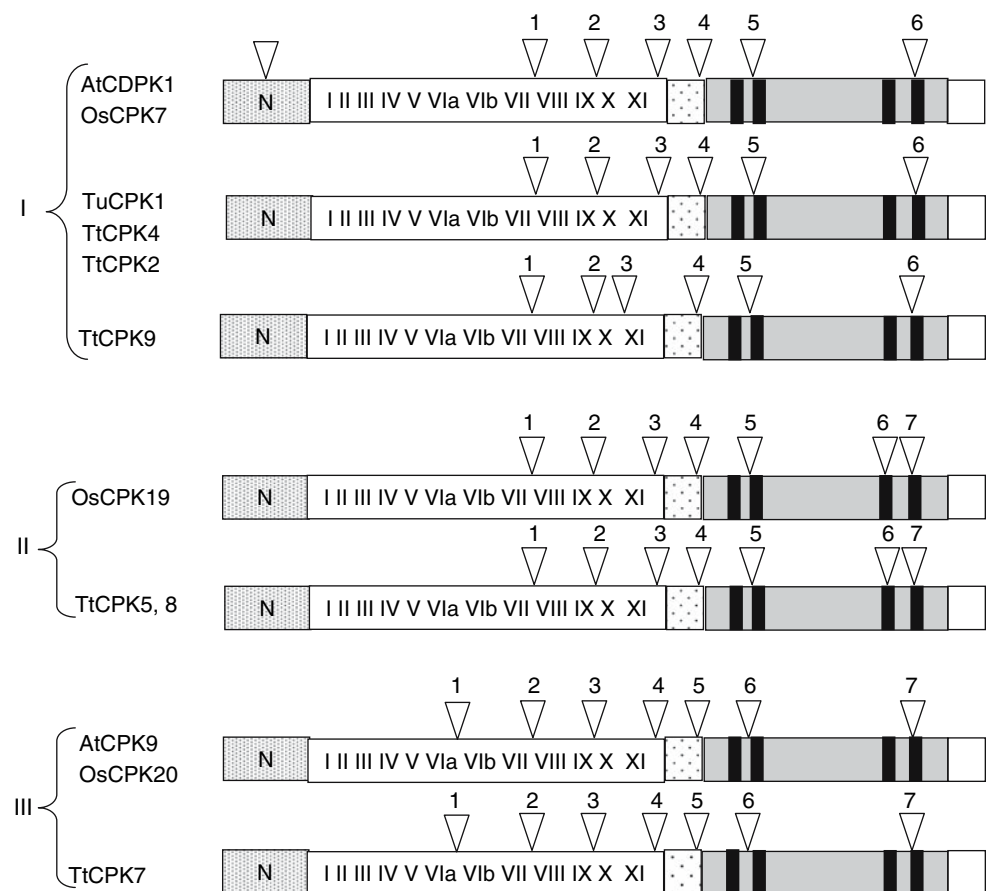


Fig. 3 Phylogenetic relationship of wheat, rice and Arabidopsis CDPKs. Neighbor-joining tree was created using MEGA3.1 program with 1,000 bootstrap sampling using full length sequences of 14 wheat, 29 rice, and 34 Arabidopsis CDPK proteins. Four groups were labeled as I, II, III, and IV. Groups II and III can be further divided into two subgroups a and b. Wheat and rice calcium and calmodulin-dependent protein kinases, TaCCaMK and OsCCaMK, were used as outgroup

(AA genome) and *T. tauschii* (DD genome) was used with primers designed from bread wheat CDPKs. As shown in Fig. 4, genes that fell in the same subgroup had the same number of introns among all three species. These introns inserted to nearly the same locations of the genes although the sizes of corresponding introns varied (Table S3). One exception was found for CDPK genes from Group I. Unlike rice and Arabidopsis CDPK genes, wheat CDPK genes TuCPK1, TtCPK2, TtCPK4 and TtCPK9 (corresponding to TaCPK1, 2, 4, and 9 respectively) seem to have lost the first intron that was located at the N-terminus in rice and Arabidopsis CDPK genes. In addition, the location of the third intron in TtCPK9 was shifted to between subdomain X and XI, instead of between the subdomain XI and the junction region (Fig. 4). These differences in genomic

Fig. 4 Genomic structures of wheat CDPK genes. Except TuCPK1 which was isolated from *T. urartu*, the genomic sequences of the remaining CDPK genes were from *T. tauschii* (prefixed with “Tt”). Intron locations relative to the protein sequences are shown by triangles. The exact sizes of introns can be found in Table S3. Dense dotted boxes: N-terminal variable domains; White boxes: protein kinase domains; I–XI, subdomains of the protein kinase domain; Thin dotted boxes: junction domains; Grey boxes: calmodulin-like domains; Black boxes: EF hands



structures indicated that wheat CDPK genes, or at least the cloned alleles, had experienced certain genomic arrangements since they diverged from rice and Arabidopsis.

Evolution history of the wheat CDPK gene family and the deduction of wheat CDPK gene number

In rice, a number of CDPK genes display high sequence similarity with each other indicating that they were derived from segmental duplication because most of them are located on different chromosomes (Asano et al. 2005), suggesting a complicated evolution history for the formation of the CDPK gene family in rice. To gain insight about wheat CDPK gene evolution, we attempted to map these genes on wheat chromosomes. We first searched our CDPK genes against the ESTs that have been mapped to wheat chromosomes by Qi et al. (2004). Five CDPKs, TaCPK2, 5, 10, 15, and 17, matched mapped ESTs on chromosomes belonging to homoeologous groups 1, 2 and 6 (Table 1). The bin map allowed locating of TaCPKs to chromosome arms (Table 1). Using Chinese Spring (CS) nulli-tetrasomic lines, we then mapped eight additional CDPKs to homoeologous groups 1, 2, 4, 5 and 6 respectively using a PCR-based approach (See Materials and Methods). The mapping

of the remaining CDPK genes was not successful. Since no evidence of gene cluster on a particular chromosome was observed, we concluded that, like in rice, wheat CDPK genes are dispersed along wheat chromosomes.

Wheat CDPKs were highly similar to their best rice matches as determined by phylogenetic analysis, with percentage identity often higher than 80% (Figs. 3, S1, and Table S2). To estimate the divergence time of wheat and rice CDPKs as well as the evolutionary history of wheat CDPK gene family, we measured the synonymous (K_s) and nonsynonymous (K_A) mutation rates between putative wheat and rice orthologous CDPK genes. Using a divergence rate of 6.5×10^{-9} mutations per synonymous site per year (Gaut et al. 1996), we subsequently estimated the divergence time of 13 pairs of wheat–rice orthologs. As shown in Table 2, the estimated time for wheat–rice CDPK orthologs are between 32–156 MYA. The divergence time of TaCPK19 and OsCPK9 was not included in the following calculation because the estimated separation time of these two genes (156 MYA) is much earlier than the known speciation time for wheat and rice (~ 50 –70 MYA). Using the remaining numbers, the average divergence time of wheat–rice orthologs was calculated at around 50 MYA with a standard deviation of 20 MYA. This rough dating gave an approximate time of wheat and rice CDPK genes

Table 2 Estimated divergence time between wheat-rice CDPK orthologs

Gene pairs	K_S	K_A	K_A/K_S	T (MYA)
TaCPK7 vs. OsCPK20	0.4137	0.0409	0.0981	32
TaCPK2 vs. OsCPK13	0.4203	0.0389	0.0926	32
TaCPK8 vs. OsCPK1	0.4881	0.0564	0.1156	38
TaCPK4 vs. OsCPK24	0.5034	0.04	0.0795	39
TaCPK6 vs. OsCPK18	0.5037	0.0777	0.1543	39
TaCPK15 vs. OsCPK3	0.519	0.0252	0.0485	40
TaCPK12 vs. OsCPK8	0.5284	0.0844	0.1598	41
TaCPK5 vs. OsCPK19	0.5561	0.0327	0.0588	43
TaCPK1 vs. OsCPK7	0.5632	0.0905	0.1606	43
TaCPK18 vs. OsCPK12	0.5735	0.091	0.1587	44
TaCPK9 vs. OsCPK11	1.046	0.1043	0.0997	80
TaCPK13 vs. OsCPK2	1.2025	0.0407	0.0339	93
TaCPK19 vs. OsCPK9	2.0343	0.0939	0.0462	156

K_S : synonymous substitution rate; K_A : nonsynonymous substitution rate; MYA: million years ago

similar to the divergence time for cereal plants (Kellogg 1998). In other words, the estimated divergence time of the majority of putative rice and wheat CDPK ortholog was largely consistent with the time when these two species diverged from the last common ancestor.

Since closely related CDPK gene pairs are frequent in the rice genome, we wondered whether such closely related gene pairs also exist in the wheat genome. When rice and wheat CDPK proteins were used together to develop a phylogenetic tree, the orthologous relationship between wheat and rice CDPK genes was clearly recognizable (Fig. S1). For two rice gene pairs, OsCPK8/20 and OsCPK12/19, each has its own closest wheat CDPK (Fig. 3). Similar to the rest of wheat–rice orthologous pairs, the divergence time between wheat–rice orthologs in these gene pairs was also close to the time of wheat–rice speciation (Table S2). The estimated divergence time for these wheat CDPK pairs was much earlier than the speciation time (>400 MYA for TaCPK5 and TaCPK18), suggesting that the duplication event leading to these CDPK genes occurred in a much ancient ancestor. Interestingly, the GC distribution of two rice CDPK genes, OsCPK12 and OsCPK19, has diverged since they arose by duplication. As shown in Fig. 2c, GC content of the first half of OsCPK19 decreased significantly when compared to its ancient duplicate OsCPK12 after the duplication event occurred. Also interesting is that such a GC content distribution pattern was maintained in their wheat orthologs TaCPK5 and TaCPK18. The estimated divergence time of the remaining six rice gene pairs (OsCPK5/13, OsCPK4/18, OsCPK2/14, OsCPK1/15, OsCPK7/23, and OsCPK11/17) indicated that these duplications also took place before rice–wheat divergence. Therefore, each of these rice CDPKs in the pair should

have its own wheat ortholog. Since we have cloned one wheat CDPK ortholog for one rice CDPK in the pair, the second rice CDPK should also have its own wheat ortholog. In other words, there should be six more wheat CDPK genes that already existed in the common ancestor of wheat and rice. Since we have cloned 20 wheat CDPK genes, the total number of wheat CDPK genes should be at least 26, close to the CDPK family size in rice and Arabidopsis (Cheng et al. 2002; Asano et al. 2005). The aforementioned wheat CDPK pair, TaCPK3/15, seems to be wheat lineage-specific because the divergence time of these two genes was dated close to the estimated speciation time of wheat and rice (53 MYA vs. 50 MYA). In rice, lineage specific CDPKs are easily distinguishable based on their high sequence similarity, such as OsCPK25 and OsCPK26, which share 99.6% identity on the protein level, suggesting that they were derived from a very recent duplication (Asano et al. 2005).

Differential expression of homoeologous CDPK genes

With a stringent condition of 95% identity over 100 bp region for sequence alignment, we investigated the numbers of ESTs representing each wheat CDPK by a BlastN search against all GenBank ESTs using CDPK sequences as queries. As shown in Fig. S2, TaCPK2 and TaCPK5 have the largest numbers of ESTs representing them, whereas CDPKs like TaCPK17 and TaCPK20 have only one. The differential expression level of wheat CDPK genes may reflect different spectra of biological functions of these genes in wheat.

The fact that the wheat genome is composed of three homoeologous genomes demands an investigation of the expression levels of each homoeologous alleles of wheat CDPKs. We then classified the ESTs that matched correspondent CDPKs using indels (insertions or deletions) and multiple SNPs (single nucleotide polymorphisms) and found that many CDPKs had more than one genotype (Table 1 and Fig. 5). Figure 5a shows the three types of EST sequences that presumably represent the transcripts from three homoeologous alleles of TaCPK2. For confirmation, we amplified TaCPK2 transcripts from wheat cultivar Chinese Spring (CS) using RT-PCR and sequenced 33 individual clones. Indeed, we found only three types of transcripts for TaCPK2 (data not shown). For in silico analysis, each type of allele should be supported by at least two ESTs. Among the 20 wheat CDPKs, 13 were found to have three homoeologous alleles (Table 1); five had two, whereas the remaining two has one. We then counted the numbers of each genotype for those CDPKs with three types of expressing alleles. The result showed that for most CDPK genes one allele was expressed at a higher level with its transcripts occupying

more than 40% of the ESTs inspected (Fig. 5b). Therefore, although we were unable to identify the genome source of each allele, the differential expression of the three CDPK alleles in bread wheat is obvious. It is worthy to mention that most indels occurred at the 5' or 3' UTRs, whereas coding regions proved to have more SNPs (Table S4). The high sequence similarity between homoeologous alleles indicated that these genes were positively selected during their evolution, apparently due to their important roles in signal transduction for stress responses.

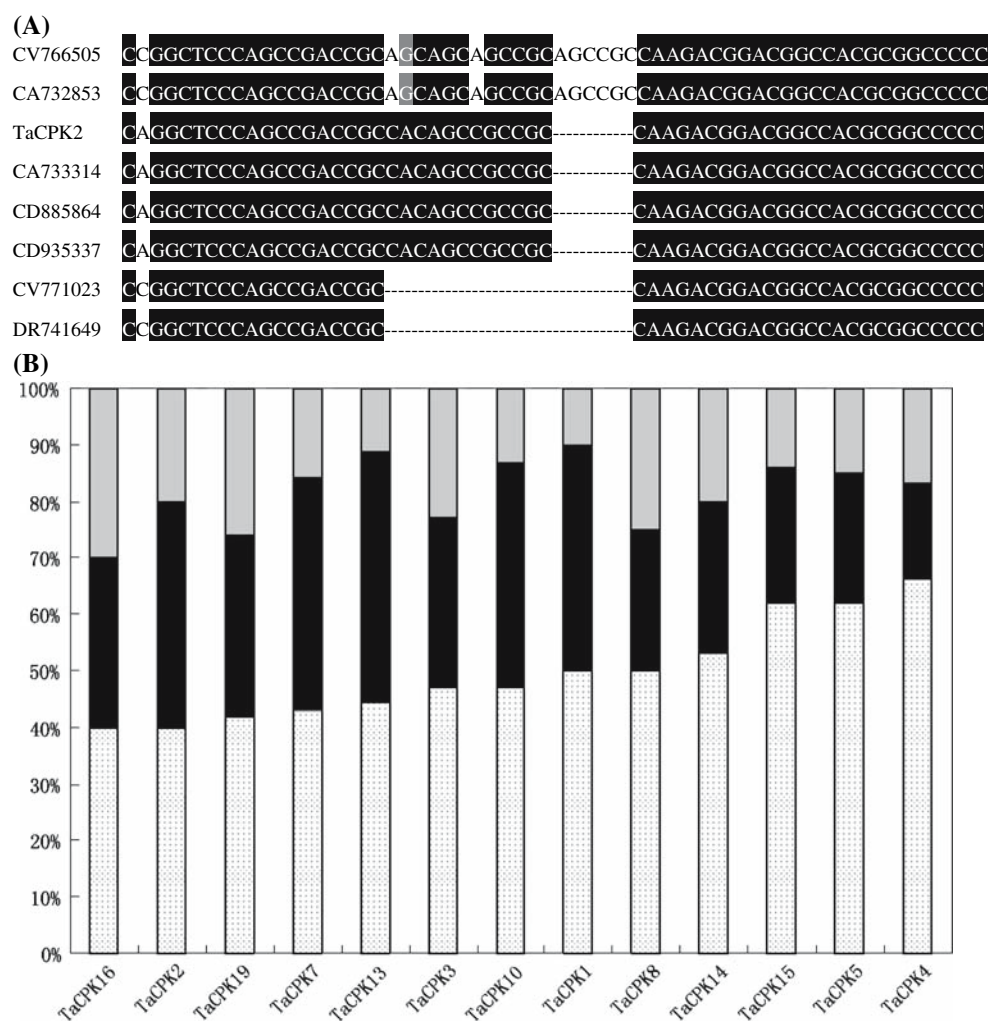
Tissue specific expression of wheat CDPK genes

CDPKs are known to be involved in various physiological processes and at different stages of plant development (Cheng et al. 2002). For example, in maize one CDPK expresses in a pollen-specific manner (Estruch et al. 1994). The expression patterns of 13 wheat CDPK genes showed five different patterns of tissue-specific expression (Fig. 6): (i) expressed in all tested tissues (TaCPK1, 2, 5, and 16); (ii)

predominantly expressed in roots, stems, leaves and young spikes (TaCPK3, 6, 9, 12, 14, and 15); (iii) predominantly expressed in roots, leaves, young spikes and immature seeds (TaCPK10); (iv) predominantly expressed in leaves and immature seeds (TaCPK8); and (v) predominantly expressed in young spikes (TaCPK13). Therefore, wheat CDPKs exhibited different tissue-specific expression patterns, suggesting that some CDPKs work in a limited number of tissues, whereas the functions of other CDPKs are broader. For the rest CDPK genes, we only tested their expression in root and leaf since these tissues are the main parts of a plant in stress responses and pathogen attacks.

Surprisingly, CDPKs of some wheat–rice gene pairs seemed to have maintained their tissue-specific expression pattern. For example, TaCPK2 and OsCPK13, TaCPK1 and OsCPK7, shared a protein sequence identity of 90.20% and 81.90% respectively (Table S2). As shown in Table S5, both wheat–rice gene pairs were widely expressed in root, leaf, and stem, suggesting that they are still playing a broad range of functions in these two species after millions of years of divergence, while other wheat–rice gene pairs

Fig. 5 Differential expression of homoeologous CDPK genes of wheat. **(a)** Polymorphisms distinguishing three types of EST sequences that presumably represent transcripts from three homoeologous alleles of TaCPK2. **(b)** The percentage of ESTs representing each genotype for 13 CDPKs with three expressing homoeologous alleles. Dotted columns indicate the homoeologous allele with the highest expression level



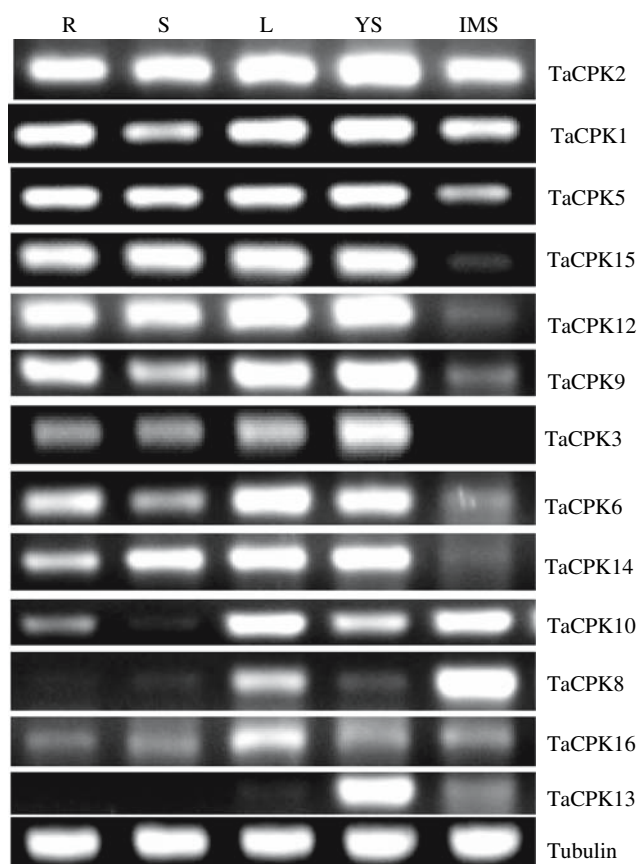


Fig. 6 Tissue-specific gene expression of 13 wheat CDPK genes. Total RNA was isolated from roots (R), stems (S), leaves (L), young spikes (YS) and immature seeds (IMS). Tubulin was used to adjust cDNA concentrations. Gene specific primers were used for semi-quantitative RT-PCR analysis of wheat CDPK genes

have changed their tissue-specific expression pattern over the course of evolution, such as TaCPK8/OsCPK1 and TaCPK4/OsCPK24 (Table S5).

Responses to biotic and abiotic stresses

One of the most important functions of CDPK genes is to sense cytosolic calcium concentration change during biotic and abiotic challenges (Klimecka and Muszyńska 2007). We therefore identified the stress and hormone responses of wheat CDPK genes. Seedling tissues were collected after being exposed to cold, drought, salt, hydrogen peroxide (H_2O_2), powdery mildew, abscisic acid (ABA), and gibberellin (GA). As shown in Fig. 7 and Table S5, 14 CDPK genes were detected by RT-PCR to respond to at least one stress treatment. Often, one CDPK was able to respond to several treatments and, on the other hand, one treatment was encountered by multiple CDPK genes. For example, nine genes (TaCPKs 1, 2, 3, 4, 7, 10, 12, 15 and 19) responded to powdery mildew treatment (Fig. 7 and Table

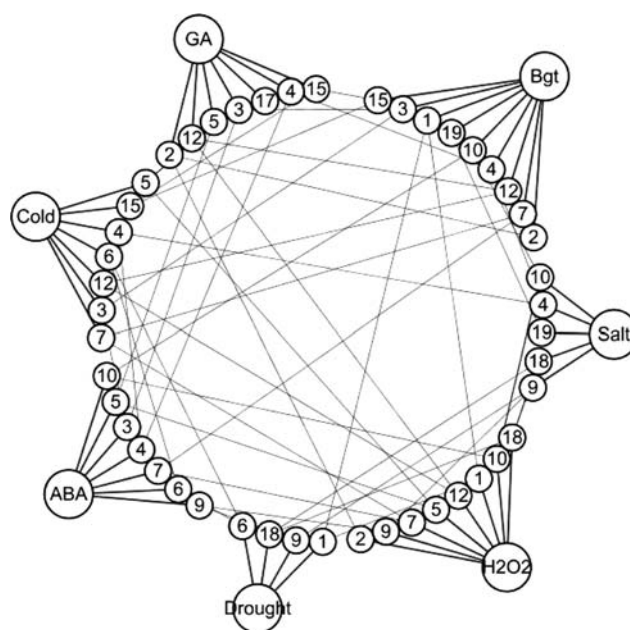


Fig. 7 Crosstalk of wheat CDPK genes in response to biotic and abiotic stresses. Abiotic stresses were performed using drought, cold, salt and H_2O_2 treatments; Wheat powdery mildew fungus *Blumeria graminis tritici* (Bgt) infection was used as a biotic challenge; Hormone stimuli included ABA and GA treatments. Numbers represent the serial numbers of wheat CDPK genes as listed in Table 1. Thick lines connecting stimuli with TaCPKs regulated them. Thin lines connect the same TaCPKs responding to multiple treatments

S6). Among these genes, for example, TaCPK4 also responded to salt, cold, ABA and GA treatments. Therefore, wheat CDPKs seem to be mediating cross talk among different signaling pathways. The stress/pathogen response conditions for the remaining CDPK genes were not available because they were not expressed in the tissues used or the expression was too low to be detected.

Again, we compared the expression patterns of wheat–rice CDPK gene pairs in stress responses. Unlike their expression in tissues, wheat CDPK TaCPK2 and its putative rice ortholog OsCPK13 responded to stresses differently (Table S5). Although both CDPKs were able to respond to pathogenic attacks, powdery mildew in wheat and blast in rice respective (Ray et al. 2007), TaCPK2 seems to have lost its ability to respond to cold, drought, and salt stresses. Therefore, as expected, the functions of these putative orthologous CDPK genes have diverged significantly since the separation of the two species.

Discussion

Conservation and divergence of the wheat CDPK gene family

CDPK genes are ubiquitous in plant genomes and are essential for many biotic and abiotic stresses, by

participating in their signal transduction pathways. In this work, we characterized this important gene family in bread wheat. Using existing wheat and barley EST information as well as our full-length cDNA libraries, we were able to isolate 20 non-redundant wheat CDPK gene fragments and subsequently obtained full-length cDNAs for 14 of these genes. In light of the fact that bread wheat is a young hexaploid species and has high sequence similarity among homoeologous alleles, full-length cDNA sequence was generated from individual clones. Global alignment showed that these CDPK genes were less than 75% identical on the DNA level except TaCPK3 and 15 (83.9%) (Table S2), whereas their identity with rice orthologs was often higher than 80%, some even close to 90%. Two wheat CDPKs, TaCPK3 and 15, were highly similar (83.9% identity on the DNA level) and seem to be wheat lineage-specific. Additional comparison of ESTs from diploid progenitors of wheat confirmed that the similarity among homoeologous copies is often higher than 95% (Zhao et al. unpublished). Therefore, we believe that these CDPK genes represent unique copies in the diploid genomes.

The structure and functional domains of wheat CDPKs were similar to those of rice and Arabidopsis. CDPKs with less than four EF hands were reported in rice and Arabidopsis, but not observed in wheat. For most wheat CDPKs, their orthologous counterparts in rice were easily distinguishable according to their sequence similarity. In rice, there are pairs of CDPK genes of high sequence similarity. Some of these gene pairs are located on two duplicated segments, whereas others are located elsewhere with little evidence for tandem duplication or segmental duplication (Asano et al. 2005). Divergence time estimation of rice CDPK gene pairs dated the occurrence of these duplication events earlier than the speciation time of rice and wheat, which is largely considered to be ~50 MYA (Ramakrishna et al. 2002). Four wheat CDPKs clearly corresponded to two pairs of rice CDPKs. These wheat CDPK pairs were dated a similar time to that of rice gene pairs, indicating their appearance predated the divergence of wheat and rice (Fig. S1 and Table S2). For additional six pairs of rice CDPKs with similar duplication time, only one wheat CDPK in each pair was cloned. We believe there should be one wheat CDPK for each of these rice CDPK gene pairs. Therefore, we estimate the diploid genome of wheat should have at least 26 CDPK genes, close to rice (31) and Arabidopsis (34). A similar number (30) of CDPKs were observed in poplar (*Populus trichocarpa* unpublished data), which has experienced three rounds of whole genome duplications (Tuskan et al. 2006). Therefore, it seems that 30 or so CDPK genes would be sufficient to mediate calcium signals conducted by CDPK proteins.

In rice, CDPKs that arose after rice diverged from wheat are easily recognizable based on their sequence similarity.

For wheat, however, only one pair of CDPKs, TaCPK3 and TaCPK15, can be considered to be wheat-specific according to their divergence time and sequence similarity. Additional wheat-specific CDPKs should exist, but will be difficult to clone using the current approach due to their high sequence similarity to the existing CDPK genes and the difficulty in distinguishing recently duplicate genes from homoeologous ones, which have only diverged for 2.5–4.5 MYA (Huang et al. 2002). With the progress of wheat genome sequencing project (<http://www.wheatgenome.org>), we anticipate that the full inventory of CDPK genes will be obtained for the hexaploid wheat.

Differential expression of CDPK genes in a polyploid species

The success of bread wheat as one of the most favorable crops for human beings should be attributed to its nature as an allopolyploid (Dubcovsky and Dvorak 2007). Although only five CDPKs were mapped to all three homoeologous chromosomes, a polymorphism survey among highly similar ESTs (>95% identity) revealed that the homoeologs of 65% (13 out of 20) of wheat CDPK genes were producing transcripts (Table 1). Five CDPKs had two homoeologous transcripts, whereas only two CDPK genes were detected single type of transcripts. Calculation of the percentage of the transcripts for each CDPK gene with three genotypes showed that for most CDPK genes, one homoeologous allele always expressed at a higher level than the remaining two. Although homoeologous gene-specific primers should be used to experimentally assess the expression level of each homoeologous gene, the counting of EST numbers did provide the first quantification about differential expression levels of homoeologous CDPK genes. Therefore, though bread wheat may have a similar number of CDPK genes for each of its diploid genome, the homoeologous alleles will certainly add another layer of complexity in terms of the gene number and the dosage effect of these signaling molecules in the hexaploid wheat.

Wheat CDPKs mediate cross-talking of multiple stress signaling pathways

Evidence is accumulating that CDPKs are involved in many physiological processes of plants (Cheng et al. 2002; Klimecka and Muszyńska 2007). However, functional study of CDPK genes is relatively limited. In rice, for example, only two CDPKs, OsCDPK7 and OsCDPK13, have been characterized using the transgenic approach (Saijo et al. 2000; Komatsu et al. 2007). Even in the model plant Arabidopsis, clear functions are only available for a few CDPK genes (Mori et al. 2006; Yu et al. 2007; Zhu et al. 2007).

Systematic analysis of wheat CDPK genes has not been reported, although a few CDPK gene sequences have been deposited in GenBank. Recently, Martínez-Noël et al. (2007) isolated a wheat CDPK (corresponding to TaCPK8 in this study) and studied its regulation by sucrose using excised leaves. We analyzed the expression patterns of all 20 CDPKs under various stress conditions. Each wheat CDPK exhibited responses to multiple stress treatments, suggesting that they were major converging points for cross talk between different signal transduction pathways. On the other hand, multiple CDPKs seemed to be necessary to coordinate with one stress stimulus. For example, nine wheat CDPKs appeared to be affected by the infection of powdery mildew, indicating the involvement of multiple signaling pathways in response to pathogen infections. The cross talk of a CDPK mediated signaling pathway with other signaling pathways has also been demonstrated. In tobacco, e.g. ectopic expression of a truncated form of NtCDPKs lacking regulatory autoinhibitor and calcium-binding domains rendered leaves to respond to an abiotic stress stimulus with activation of biotic stress reactions (Ludwig et al. 2005). These responses included synthesis of reactive oxygen species, defense gene induction, and SGT1-dependent cell death. Furthermore, elevated CDPK signaling alleviated stress-induced MAPK activation, a process requiring ethylene synthesis and perception. Such a cross talk suggested that CDPK and MAPK pathways do not function independently and that a coordinated activation of both pathways are needed for efficient responses to biotic and abiotic stress (Ludwig et al. 2005). Nevertheless, how one CDPK mediates multiple signal transduction pathways and in the meantime maintains its substrate specificity remains to be elucidated. For wheat, functional study of CDPK genes is just beginning. Extensive study of wheat CDPKs will prove to be important for understanding the molecular mechanisms of wheat stress responses and aid in breeding enhanced wheat cultivars for agriculture.

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